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$$(4, 6, 6, 4, 0, -6, -14, \dots)$$

$$\Rightarrow (2, 0, -2, -4, -6, -8, \dots)$$

$$\Rightarrow (-2, -2, -2, -2, -2, \dots)$$

$$x_n = an^n + bn + c$$

$$\Rightarrow \begin{cases} a + b + c &= 4 \\ 4a + 2b + c &= 6 \\ 9a + 3b + c &= 6 \end{cases} \Rightarrow \begin{cases} 3a + b &= 2 \\ 5a + b &= 0 \end{cases} \Rightarrow 2a = -2 \Rightarrow a = -1$$

$$\Rightarrow b = 5 \Rightarrow c = 0$$

$$x_n = -n^2 + 5n = n(-n + 5)$$

$$x_8 = 8(-8 + 5) = 8(-3) = -24$$

References